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Thermal remote sensing of water under flooded vegetation: New observations of inundation patterns for the 'Small' Lake Chad

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SUMMARY

Lake Chad at the border of the Sahara desert in central Africa, is well known for its high sensitivity to hydroclimatic events. Gaps in in situ data have so far prevented a full assessment of the response of Lake Chad to the ongoing prolonged drought that started in the second half of the 20th century. Like many other wetlands and shallow lakes, the 'Small' Lake Chad includes large areas of water under aquatic vegetation which needs to be accounted for to obtain the total inundated area. In this paper, a methodology is proposed that uses Meteosat thermal maximum composite data (Tmax) to account for water covered by aquatic vegetation and provide a consistent monthly time series of total inundated area estimates for Lake Chad. Total inundation patterns in Lake Chad were reconstructed for a 15-yr period (1986–2001) which includes the peak of the drought (86–91) and therefore provides new observations on the hydrological functioning of the 'Small' Lake Chad. During the study period, Lake Chad remained below 16,400 km^2 (third quartile ~8800 km^2). The variability of the inundated area observed in the northern pool (standard deviation $\sigma_{\text{northern pool}}$ = 1980 km^2) is about 60% greater than that of the southern pool ($\sigma_{\text{southern pool}}$ = 1250 km^2). The same methodology could be applied to other large wetlands and shallow lakes in semi-arid or arid regions elsewehere using Meteosat (e.g. Niger Inland Delta, Sudd in Sudan, Okavango Delta) and other weather satellites (e.g., floodplains of the Lake Eyre Basin in Australia and Andean Altiplano Lakes in South America).

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1. Introduction

Across the world most shallow lakes and wetlands are largely or in part covered by aquatic vegetation (Burgis and Symoens, 1988; Hughes and Hughes, 1992; Talling and Lemoalle, 1998; Fraser and Keddy, 2005). In central Africa, the transition from an 'Average' to a 'Small' Lake Chad in 1973 (Figs. 1 and 2) was accompanied by the development of aquatic vegetation which now dominates the inundated area of Lake Chad (Carmouze et al., 1983; Lemoalle, 2004). Despite the critical need for information during stages of low levels, the hydrological functioning of the 'Small' Lake Chad during drought periods is not well documented (Olivry et al., 1996; Lemoalle, 2004). While the gauging station in Bol has been operating almost continuously since 1953 in the southern pool (Fig. 2), no in situ level data are available from the northern pool of the lake since 1975, when the lake was split into different water bodies. Quantitative long-term time series of observations of the northern pool can only originate from satellite data.

Using the earliest data available from satellite radar altimetry Topex/Poseidon (T/P), it is possible to estimate water height fluctuations in the southern pool as far back as 1992 (Birkett, 1995, 2000; Cretaux and Birkett, 2006). T/P radar altimeter has no track above the northern pool but ERS-2 and Envisat satellites could provide some water levels from 1995 and 2002 respectively. Radar altimetry in the northern pool remains difficult given the complex nature of this region with the presence of islands, small channels, and changing vegetation.

Satellite data have been used extensively to measure inundated areas, which is arguably a more pertinent parameter than water level to capture the dynamics of large shallow water bodies (e.g., Ozesmi and Bauer, 2002; Jones et al., 2009). However, main limitations still remain; especially for the detection of water under flooded vegetation. Optical sensors, such as Landsat TM and MODIS, can easily detect open water using the strong absorption of solar energy by water in the near and middle infrared. Shallow depths and turbid waters, are better detected at greater wavelengths (>1 μm; mid-infrared/short-wave infrared) where the illumination of the suspended materials or of the shallow bottom of a water column is considerably reduced (Li et al., 2003; Bukata, 2005). The mapping of water underneath vegetation

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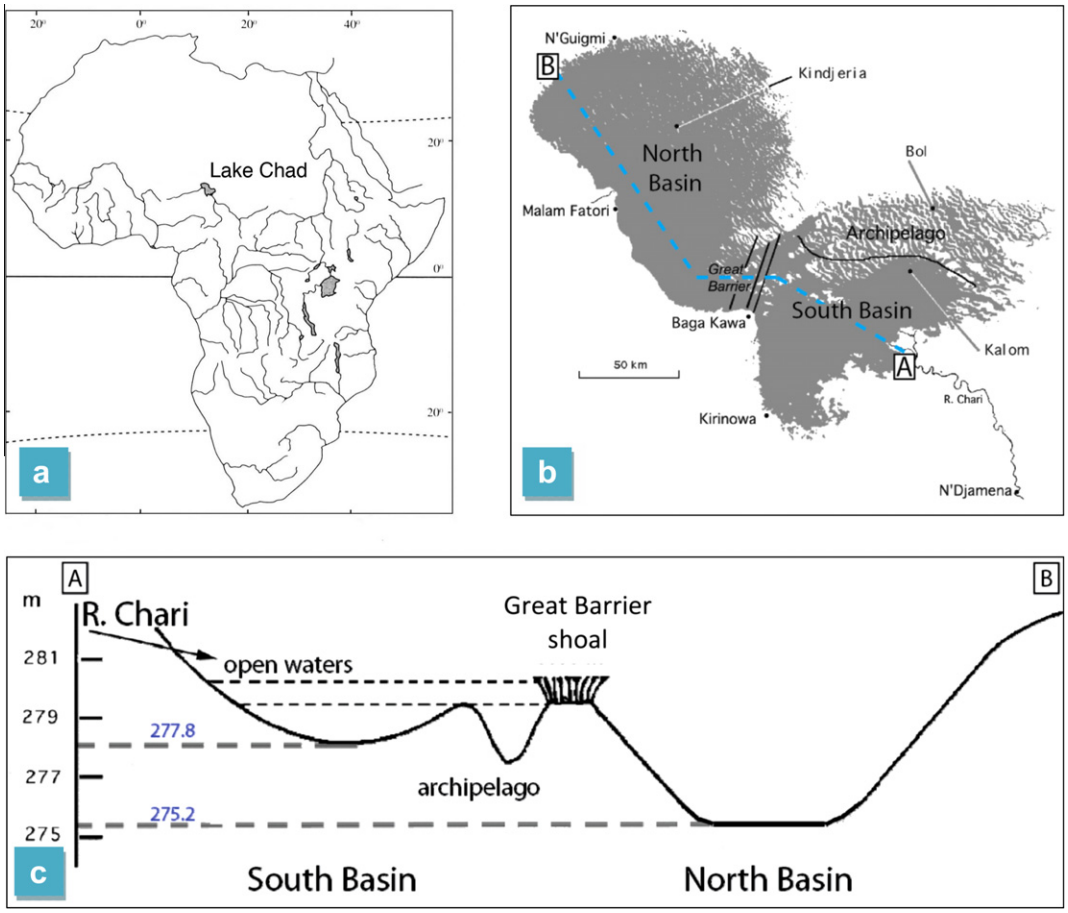

Fig. 1. Lake Chad: (a) location of Lake Chad in Africa; (b) the extent of the 'Average' Lake Chad from a Landsat mosaic of January 1973 with the main shoals between the pools. The gauging stations are indicated, with Bol still being operational; and (c) a schematic cross section of Lake Chad showing the two main pools, with indications of the elevation (m amsl) of the sills and pool beds (modified from Lemoalle et al., 2008).

is problematic for optical sensors as they only observe the skin surface of the vegetation. This limitation may be overcome by using classification techniques with multispectral data to selectively map hydrophyte plants (e.g. Zomer et al., 2009). However hydrophyte plants may survive in wet soils when the inundation receeds and other types of vegetation (e.g. forest) may occasionally get flooded. In water limited environments, a high photosynthetic activity will differentiate flooded vegetation from the rest of the dry land. Satellite based vegetation indices, which are derived from optical data, give a measure of the photosynthetic activity. Vegetation indices from the NOAH/AVHRR and MODIS sensors have been used in semi-arid and arid regions to map water under aquatic vegetation across large wetlands outside the rainy season (e.g., Leblanc et al., 2003b; Mariko, 2003; Sakamoto et al., 2007). This technique is hindered by the fact that vegetation activity may often remain high for several weeks after water has receded. Normalised Difference Wetness Index (Gao, 1996; Xiao et al., 2002), or the Tasseled Cap wetness index (Ordoyne and Friedl, 2008) have also been applied. However the main limitation of all these optical-based approaches is the inability to map beneath clouds; and this is particularly a problem for tropical regions which are characterised by frequent cloud coverage.

Radar imaging is less affected by cloud cover and can penetrate vegetation at a depth which depends on the wavelength used and the structure (density and height) of the vegetation. Synthetic Aperture Radar (SAR) sensors are active instruments that measure the backscattering coefficient of target surfaces (e.g., ERS1 and 2, Envisat ASAR, JERS, PALSAR, RADARSAT). Under low wind conditions, open water reflects specularly the radar signal resulting in a characteristic low backscatter (return) to the sensor, while water underneath vegetation may often be identifiable by a double bounce effect (Lewis, 1998). A main limitation of radar data is that wind-induced waves (Braggs waves) increase scattering from open water surfaces (Alpers, 1985; Smith and Alsdorf, 1998). As a result the radar signal over water surfaces continually changes with waves. This phenomenon is particularly strong in C band but is often insufficient to affect L band data (Smith and Alsdorf, 1998; Alsdorf et al., 2007). L band is also capable of greater penetration of the vegetation and has been used successfully to map flooded areas under thick forest (Hess et al., 2003; Frappart et al., 2005; Martinez and Le Toan, 2007; Rosenqvist et al., 2007; Alsdorf et al., 2007). The use of L band data from the JERS and PALSAR sensors for inundation monitoring is mainly restricted by acquisition times and limited archives rather than by weather or vegetation condition. In the future L band archives may improve with the recent launch of a second PALSAR satellite and more systematic acquisitions, but, in general, frequent acquisition of radar data over large areas may remain expensive and impractical. Passive microwave data (e.g. SSM/I and ISCCP) are helpful for delineating inundated areas (e.g., Sippel et al., 1998; Hamilton et al., 2002), in particular when used in conjunction with other sensors to limit confounding factors such as atmospheric condition and vegetation (Prigent et al., 2007), but their use is limited by their low spatial resolution (tens of km).

To date, the remote sensing of inundation patterns in the specific case of Lake Chad can be summarised as follow. Mohler

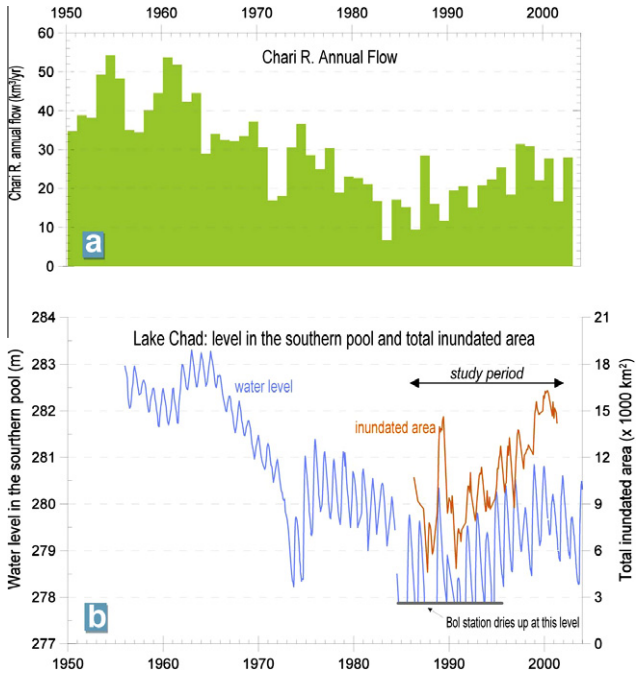

Fig. 2. Hydrological fluctuations in Lake Chad and the Chari Basin: (a) annual flow of the Chari River at N'Djamena in km^3 and (b) Lake Chad level in the southern pool and total inundated area. In situ lake level data are from Kiri station from February 1979 to May 1984 and Bol station for the rest of the time. Satellite estimates of total inundated area are from this study.

et al. (1989) used registered photographs from manned space flights to estimate the open water area of Lake Chad for 8 dates between 1966 and 1986. Citeau et al. (1989) and Rigal (1989) used Meteosat channel 1 in the visible domain (2.5 km resolution) to monitor the dynamics of open water in 88/89. Wald (1990) compiled a time series showing the variation of the area of open water in Lake Chad between 1966 and 1989 (23 dates with 83% of the data between 1984 and 1989) derived from space photography (Mohler et al., 1989; Wood et al., 1989) and Meteosat visible channel (Anonymous, 1989; Citeau et al., 1989; Rigal, 1989; Wald, 1990). The area of the annual maximum inundation in the northern pool observed in January was derived using Landsat MSS band 7 (0.8–1.1 μm) data from 1973 to 1976 and Meteosat data in the visible channel from 1977 to 1990 (Lemoalle, 1978, 1991; Olivry et al., 1996). More frequent inundation maps capturing the seasonal variability of Lake Chad from 1995 to 1998 were obtained by Birkett (2000) using a time series of NOAA/AVHRR LAC images. Birkett (2000) used a threshold technique in AVHRR band 2 (0.7–1 μm) to detect open water. Rigal (1989) first demonstrated the use of thermal infrared data to map inundated areas of Lake Chad under aquatic vegetation. The author used nine AVHRR images between November 1988 and April 1989 to estimate the variations of the total inundated area of Lake Chad, using a normalised water index based on the contrast of water (open and covered by vegetation) in the thermal infrared (band 5) and near infrared (band 2) channel. Rosema and Fiselier (1990) used the principle of thermal inertia to detect the total inundated area using Meteosat noon and midnight thermal images in the Yaeres flood plain, south of Lake Chad which are typically covered by tall grasses. They captured both open water and water covered by aquatic vegetation but processed only 11 dates between 1978 and 1986. Travaglia et al. (1995) also applied this thermal inertia method using AVHRR/LAC thermal data (band 4) to map the extent of water bodies in the Sudd wetlands in Sudan. This region has a similar environment to that of Lake Chad with wetlands covered by dense mats of floating vegetation within which areas of open water occur. Travaglia et al. (1995) report that AVHRR/LAC thermal data (band 4) can achieve a clear distinction of both open water and water under vegetation from the bare soil and soil with active vegetation.

In this paper, a methodology is proposed that uses Meteosat maximum thermal composite data (Tmax) to account for water under flooded vegetation and provide a consistent time series of monthly estimates of the total inundated area of Lake Chad. The long archive of Meteosat thermal composite data also allows us to reconstruct monthly inundation patterns of the lake from 1986 to 2001 and therefore provides new observations on the hydrological functioning of the 'Small' Lake Chad.

2. Study area

2.1. Lake Chad variability and landscapes

Lake Chad lies in an endoreic basin at the centre of Africa on the southern margin of the Sahara in the semi-arid Sahel belt. It is a highly variable shallow lake centred approximately on 13°30′N and 14°E (Fig. 1). The lake belongs to Chad, Niger, Nigeria and Cameroon. It provides a wealth of natural resources (fisheries, recession cultivation and rangeland) which are highly attractive to many populations in the region, especially during drought years.

Alike most closed shallow lakes, Lake Chad has a long history of wet and dry periods which span different time scales, from the Quaternary to the century (Servant and Servant, 1983; Olivry et al., 1996; Lemoalle, 2004; Leblanc et al., 2006a,b). As a result of rainfall fluctuations over the basin of the Chari River, during the past 150 years Lake Chad has been alternatively described as a large inland sea, a large lake with numerous islands or a great marshy area. Tilho (1928) proposed the denomination of 'Large', 'Average' and 'Small' Lake Chad for the three main states of the lake, according to its water level, areal extent and landscape. A summary of the main characteristics for the different states of Lake Chad is given in Table 1 (Lemoalle et al., 2011). A 'Large' Lake Chad, with a water level around 284 m amsl has been observed by European explorers who visited the lake in the 19th century. The 'Average' Lake Chad has an area of about 18,000–22,000 km^2 for a water level of 281–282 m, whilst the 'Small' Lake Chad has an area fluctuating between 3000 and 14,000 km^2 and, below 279 m, is divided into two separate large pools with differing water levels (Fig. 1b). The pools of the 'Small' Lake Chad are separated by shoals covered with dense vegetation through which the water overflows only occasionally during flooding events. One of the main shoals is the 'Grande Barrière' or 'Great Barrier' (Fig. 1b). In these conditions of a 'Small' Lake Chad, several water level gauges are needed to describe the lake status in the respective pools. A 'Small' Lake Chad is also characterised by the spread of aquatic vegetation mainly composed of rooted (*Phragmites australis*, *Cyperus papyrus*, *Vossia cuspidata*, *Aeschynomene elaphroxylon*) and floating hydrophytes (*Pistia stratiotes* and *Salvinia molesta*) which covers most of the inundated area (Iltis and Lemoalle, 1983; Lemoalle, 2004; see also Fig. 11c).

Table 1
The main characteristics of the different states of Lake Chad.

Lake Chad	'Small'	'Average'	'Large'
Chari discharge (km^3/yr)	10–34	40	45
Water level (asl)	Several levels	279–282	>282.3
Number of large water bodies	Several	One	One
Total lake area (km^2)	2000–14,000	18,000–22,000	24,000
North basin area (km^2)	0–7000	9000	10,000
Landscape	Marshes	Dune islands	Open waters
Aquatic vegetation	+++	++	+

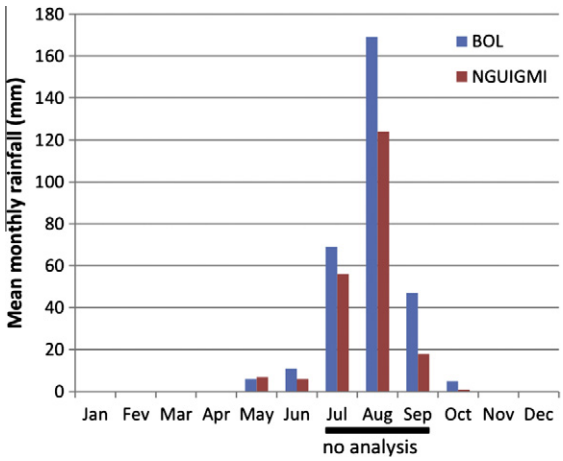

Fig. 3. Mean monthly rainfall over Lake Chad recorded at two stations in the northern (N'Guigmi: 1922–1976) and southern basins (Bol: 1932–1977). Adapted from Olivry et al. (1996).

Table 2
Mean values for the components of the water budget of an 'Average' Lake Chad (1954–1969). Values expressed as water height (mm/yr) over the lake surface area (from Vuillaume, 1981).

Components of the water budget for the 'Average' Lake Chad	Annual fluxes (mm/yr)
River inputs	+1946
Direct rainfall	+329
Evaporation	−2170
Net seepage	−102

2.2. An asynchronous annual flood

The climate in the lake region is characterised by a short rainy season from June to October and a pronounced dry season for the rest of the year (Fig. 3; Olivry et al., 1996). The annual flood of the lake does not depend on the local rainfall but on the slow transfer of flood waters generated from heavier rainfall events in the southern part of the basin. For this reason, the annual flooding of Lake Chad peaks in January in the middle of the local dry season (Fig. 3).

2.3. Lake Chad water budget

Except during Large Lake Chad periods, the lake has no surface outflow. River and rainfall inputs are offset by evaporation losses, some seepage and by volume and area fluctuations according both to seasonal and annual variations of the inflows. The water budget of Lake Chad has been studied by a number of authors (Olivry et al., 1996). Mean water budget values for the 'Average' Lake Chad during the 1954–1969 period are given in Table 2. The main input to the lake is the discharge of the Chari River (41.7 km^3/yr or 82.3% of total inputs), supplemented by direct rainfall (14% of total inputs) and inflow from the other small tributaries, El Beid and Komadugu Yobe (1.43 and 0.49 km^3/yr respectively or 3.6% of total inputs). The losses result mainly from evaporation (95.5%) and seepage out of the lake (4.5%). The seepage may however change in direction, seasonally or from year to year, according to the trend in the change of the water level.

The hydrology of the 'Small' Lake Chad is controlled by the Great Barrier which extends more than 40 km between the two main basins and is covered with a dense vegetation acting as a break to the circulation of water. Since 1973, as a consequence of the drought in the African Sahel belt, the 'Great Barrier' has been separating the southern pool of the lake, fed directly by the inflows from the Chari River (annual mean 19.4 km^3/yr between 1983 and 2000), from the northern pool which receives only a small contribution from the Yobe River (annual mean 0.44 km^3/yr between 1983 and 2000) and the overspill of the southern basin when the inflow of the Chari River is sufficient (Fig. 1c). In the southern pool, starting from a seasonal low level in July or August, an increase in water level occurs with the arrival of the Chari River flood, until a maximum water level is reached in December. If a water level of 279.5–280 m is reached, some overflow occurs toward the northern basin of the lake through the Great Barrier. An annual Chari discharge exceeding about 15 km^3/yr is needed for this overflow to occur (Bader et al., 2011). The water budget of the 'Small' Lake Chad is quite complicated because of the separate behaviour of the different basins. A hydrological model has been developed to simulate the water level and extent in these basins (Bader et al., 2011). An objective of this paper is to provide monthly observations of the 'Small' Lake Chad for the calibration and validation of this model.

3. Data and methods

3.1. Method rationale and background

Land Surface Temperature may be used as an indicator of relative soil moisture and surface water distribution (Idso et al., 1975; Byrne et al., 1979; Sabins, 1999; Leblanc et al., 2003a). Firstly, water has a higher thermal inertia than land, so that during the day/night cycle a dry land surface generally heats and cools faster than a wet surface, and to a greater degree. Secondly, in large water bodies, heat may also be distributed through mixing. Thirdly, surface water bodies have an 'unlimited' supply of water for evaporation, and an effect of evaporation is the cooling of the surface affected. In the semi-arid region of Lake Chad, a strong discrepancy between diurnal and nocturnal temperature is observed all year long (Olivry et al., 1996). These conditions of high daily temperature range enhance the thermal contrast between water and the land through the phenomena described above and are favourable to the use of thermal satellite data for the mapping of inundated areas.

3.2. Meteosat Tmax data

30-day Meteosat Tmax data were generated by Meteo-France and the Research Institute for Development (IRD) in the framework of the" "Veille Climatique Satellitaire" programme (1985–2001). Meteosat is a geosynchronous satellite which recorded the brightness temperature of the land surface by means of its thermal infrared band (10.5–12.5 μm) for the whole of Africa and a large part of the world with a temporal resolution of 30 min. 30-day Tmax data are processed as the mean of maximum value composite (MVC) of all calibrated brightness temperatures obtained using the sensor's thermal infrared band over six 5-day spans. The method has the advantage of bypassing the clouds and reducing atmospheric effects whilst maintaining a relatively good memory of surface events (Lahuec and Guillot, 1994). The compositing of thermal images in Tmax data is a multi-temporal signature of the surface characteristics and meteorological events. Tmax data have been used to monitor the agricultural (wet) season in the African Sahel domain by providing a long-term comparable indicator of soil moisture (Lahuec and Guillot, 1994; Leblanc et al., 2003a). Tmax data do not have atmospheric correction and emissivity correction applied to them; this was acceptable here as we are interested in

relative variations in the surface temperature rather than absolute temperature.

There are several advantages of using Tmax data for monitoring the fluctuations of Lake Chad. Firstly, a long archival record of Tmax data is available from May 1986 to July 2001, with only 1 gap in November and December 1986. Thus, Tmax archives cover most of the drought period that has affected Lake Chad in the second half of the 20th century. Secondly, the high temporal resolution of Meteosat allows for effective bypassing of clouds. For instance during the study period, Meteosat has a much higher temporal resolution (30 min) than AVHRR (twice daily). This high temporal resolution also increases the chances to get closer to the actual maximum surface temperature of the day and have consistent data. Thirdly, Meteosat has a relevant viewing area and spatial resolution (5 km) for the monitoring of Lake Chad ($\times$1000 km^2).

3.3. Retrieval of Tmax inundated data

Tmax archive data were processed to reconstruct a time series of total inundation of Lake Chad from 1986 to 2001. Inside the shoreline of the 'Large' Lake Chad we can assume that, during the dry season, consistently cold pixels (low Tmax values) relate to inundated areas, while warmer pixels (higher Tmax) correspond to dry areas (refer to Sections 3.1 and 3.2 for rationale). The use of a constant thermal threshold value to isolate inundated from dry pixels was rejected because (i) Meteosat Tmax data do not have atmospheric and emissivity corrections applied to them and (ii) we sought relative rather than absolute temperature differences. Instead, we used an unsupervised classification to automatically separate two natural clusters (relatively low Tmax/inundated and higher Tmax/dry pixels). Unsupervised classification algorithms essentially examine a large number of unknown pixels and divide them into a number of classes based on natural groupings according to their statistical distribution (Richard and Jia, 2006). Fig. 4a shows an example of the total inundated area detected in the Meteosat Tmax data using the unsupervised classification in June 1999. Inundated areas in red, as delimited from the unsupervised classification, clearly stand out as a group with a distinctively cooler monthly Tmax signature than the rest of the landscape inside the Lake Chad shoreline (green[1] boundary, Fig. 4a). The histograms showing the frequency distribution of the monthly Tmax values inside the boundary of Lake Chad for the corresponding period also shows two distinct clusters (inundated and dry pixels) in the Tmax data (Fig. 4b).

3.4. Validation of Tmax inundated areas

Difficulties in accessing the study area, and in particular the northern pool of Lake Chad, severely limit the possibilities of in situ validation of the inundation estimates from the Tmax data. We visited the northern pool of Lake Chad in June 1999 and recorded the GPS position of the shoreline of a very large and connected inundated area (reported in blue in Figs. 4, 5,11a and 11b; photo 1). This surveyed limit agrees well with the boundary of the inundated area detected in the corresponding Meteosat Tmax data (Fig. 4).

A comparison of the inundated areas mapped in June 1999 was also performed across sensors between AVHRR/LAC (08/06/1999 13:45 pm UT) and Meteosat (Tmax June 1999) (Fig. 5). A simple threshold technique on the AVHRR near infrared band 2 (Birkett, 2000) only enables the detection of open water ($\sim$3500 km^2, Fig. 5a). But the AVHRR NDVI (Fig. 5b) shows that the vegetation around this open water is very active (bright pixels). Such active vegetation at the end of the dry season in a semi-arid to arid region can only be sustained by the lake water; this is aquatic vegetation and covers an estimated $\sim$10,900 km^2 in Fig. 5b. Hence a total inundated area (open water and water under aquatic vegetation) of $\sim$14,400 km^2 in June 1999 according to AVHRR near infrared and NDVI data. The two thermal datasets (AVHRR band 4 in Fig. 5c and Meteosat Tmax in Fig. 5d) detect both the open water and water under aquatic vegetation as cooler areas of the landscape in dark pixels (Leblanc et al., 2003b, 2007). Note the similarity of the inundated area observed by AVHRR and Meteosat Tmax thermal data. Applying a two clusters unsupervised classification to the AVHRR thermal infrared data (Fig. 5c) returns a total inundated area of 15,500 km^2 in June 1999; an area only $\sim$5% larger than the 14,800 km^2 total inundated area detected with the Meteosat Tmax.

4. Results

We found the applicability of the Meteosat thermal mapping technique is limited during the local rainy season, when the wet soils around the lake will have a similar thermal signature to the inundated areas. We therefore had to exclude most of the local rainy season from the analysis; all July, August and September and $\sim$60% of the June and October Meteosat Tmax data were discarded. However, this still leaves us with up to 9 months of valid data per year and a data set that is perfectly timed to capture the annual flooding of Lake Chad centred in January (Table 3).

All the monthly Tmax-derived inundation areas for the 1986–2001 study period are presented in Fig. 6 for the entire lake and the northern pool. Fig. 6 also shows the average monthly discharge of the Chari River. The peak of the annual lake cycle is not centred on the wet season (JAS), but is controlled by the arrival of the Chari annual inflow with a time lag as water propagates in the southern and then the northern pools during the dry season. Observations during the entire period shown an average time lag between the annual peak flow of the Chari River at N'Djamena and the arrival of the peak inundation in the northern pool is 4 months; instead of 2 to 3 months for an 'Average' Lake Chad (water level data from 1967 to 1971). The entire lake exhibits seasonal fluctuations in inundation areas between $\sim$500 km^2 in 00/01 and $\sim$8200 km^2 in 88/89 (average seasonal range $\sim$3000 km^2). The north pool has seasonal fluctuations ranging from $\sim$0 km^2 (in 86/87; 87/88; 89/90; 90/91 and 93/94) to $\sim$4800 km^2 in 88/89 (average seasonal range $\sim$1500 km^2). An example of the fluctuations in the inundated area of Lake Chad, as mapped using the monthly Tmax data, is shown in Fig. 7 when the 88/89 annual inflow from the Chari River (28 km^3) reached the lake at its driest and resulted in the largest seasonal increase in inundated area.

Accordingly, two annual hydrological cycles have been identified for the northern pool of the 'Small' Lake:

1. An almost complete dryness the whole year in the absence of inflows from the southern pool. Only the mouth of the Komadugu Yobe River is periodically inundated during 4 to 5 months.
2. The arrival of a flood from the southern pool and through the 'Great Barrier' in December with a peak of inundation in March followed by a partial or total recession. The large season fluctuation of the inundated area during this cycle is beneficial to recession cultivation and livestock.

Over the study period (1986–2001), the entire lake showed an inter-annual variability ranging from a minimum inundated area of $\sim$4600 km^2 in October 87 to a maximum of $\sim$16,300 km^2 in

[1] For interpretation of color in Figs. 1–11, the reader is referred to the web version of this article.

February 00. Whilst the northern pool varied from being completely dry in October 86, October 87–May 88 and October 90, to having a maximum inundated area of ~6000 km^2 in February 2000.

From 1986 to 2001 the inundated area of the entire lake has a first quartile (25th percentile) of ~8800 km^2 and third quartile (75th percentile) of ~13,700 km^2 whilst the inundated area of the northern pool has a first quartile of ~360 km^2 and third quartile of ~3500 km^2. In Fig. 8, the inundation patterns of Lake Chad are expressed as inundated area duration curves for the northern pool and the entire lake. Over the study period, the northern pool was completely dry for 7% of the time (plus July, August, September) and it exceeded 4500 km^2 (35% of its total area as defined by the 284 m amsl shoreline) for only about 20% of the time.

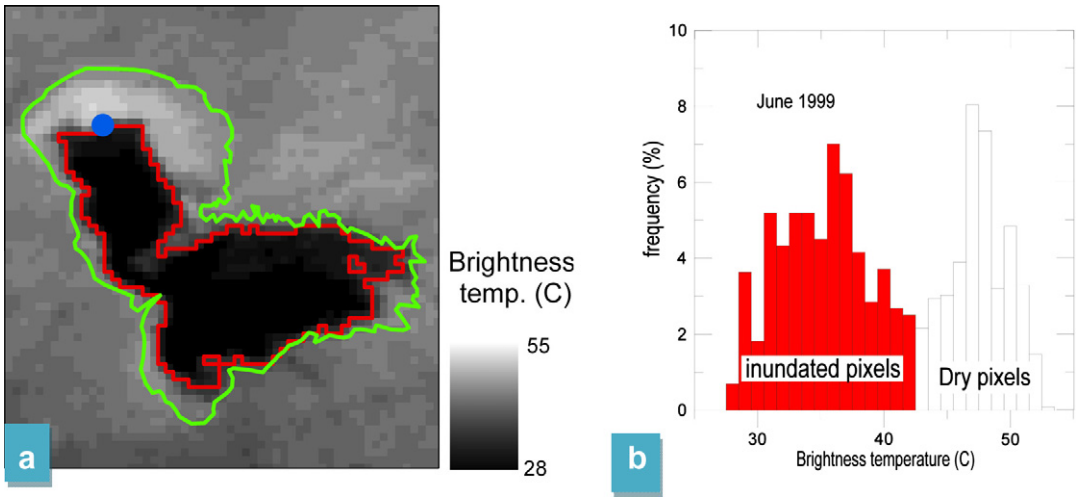

Fig. 4. Example of the detection of the total inundated area (open water and water under aquatic vegetation) using an unsupervised classification of the Meteosat maximum temperature composites (Tmax) in June 1999. (a) Monthly Tmax image of the Lake Chad region in June 1999 with the total inundated area detected using the unsupervised classification delimited in red and the shoreline of the 'Large' Lake Chad in green. The point in blue indicates the shoreline of Lake Chad observed in June 1999 in the vicinity of Doro fishing village 14°04′48″N–13°29′46″E (see also photo 1). And (b) histogram showing the distribution of the brightness temperatures of the monthly Tmax pixels inside the 'Large' Lake Chad shoreline with the inundated pixels detected using the unsupervised classification in red.

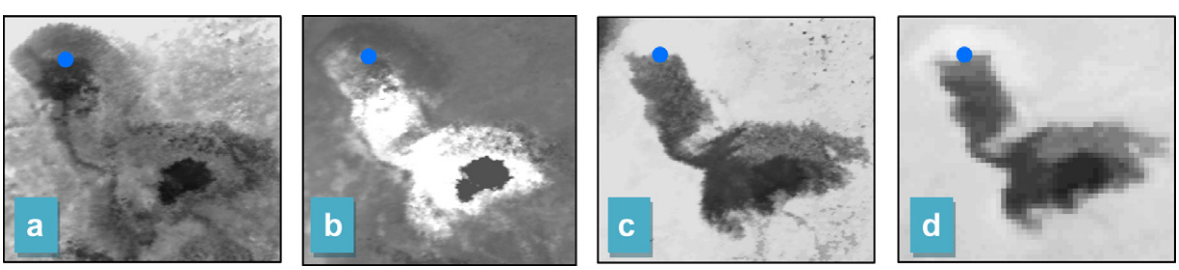

Fig. 5. Comparison of Lake Chad as observed in June 1999 with AVHRR (08 June 1999; 13:45 UT) and Meteosat thermal infrared maximum composite (Tmax 1–10 June 1999): (a) AVHRR near infrared band 2 only open water is detected in dark tones; (b) AVHRR normalised difference vegetation index – active aquatic vegetation appears in bright tones; (c) AVHRR thermal infrared – both water and aquatic vegetation which remains cooler than dry land during the day appear in darker tones; and (d) Meteosat thermal infrared maximum composite (Tmax) both water and aquatic vegetation which remains cooler than dry land during the day appear in darker tones. The point in blue indicates the shoreline of Lake Chad observed in June 1999 in the vicinity of Doro fishing village 14°04′48″N–13°29′46″E (see also photo 1).

Table 3
Calendar showing monthly Meteosat Tmax data analysed; with inundated detected for entire lake (×100 km^2). No data were analysed during the wet season (July–September).

Year	January	February	March	April	May	June	//	October	November	December
1986					107			92		
1987					87			46	67	78
1988	76	74	65	58	60			84	99	127
1989	140	136	138	137	144	146			84	88
1990	94	92	91	84	95	88		48	62	71
1991	74	71	72	66	73			78	86	95
1992	101	101	98	110	96			79	84	91
1993	97	103	108	110	109	114			85	86
1994	98	89	88	94	87				94	98
1995	102	113	113	119	121	127			98	103
1996	104	119	127	121	133				88	0
1997	117	124	134	138	137				113	122
1998	122	123	125	125	128				122	136
1999	148	151	156	151	150	148			150	160
2000	158	163	161	162	163	162			147	156
2001	145	152	150	149	142					

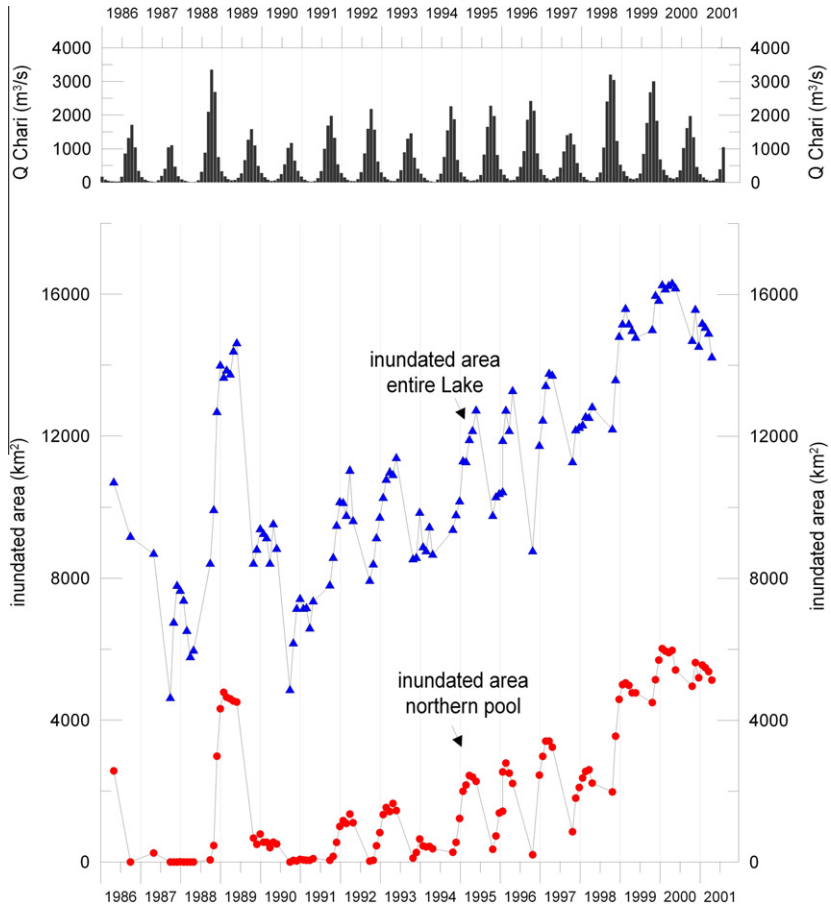

Fig. 6. Graph showing monthly mean discharge of the Chari River at N'Djamena and monthly fluctuations of the total inundated area of Lake Chad (blue line) and the northern pool (red line) as derived from Meteosat Tmax data from May 1986 to May 2001. Discharge data from the Chari River at N'Djamena are from the Direction des Ressources en Eau et de la Météorologie (DREM, Chad).

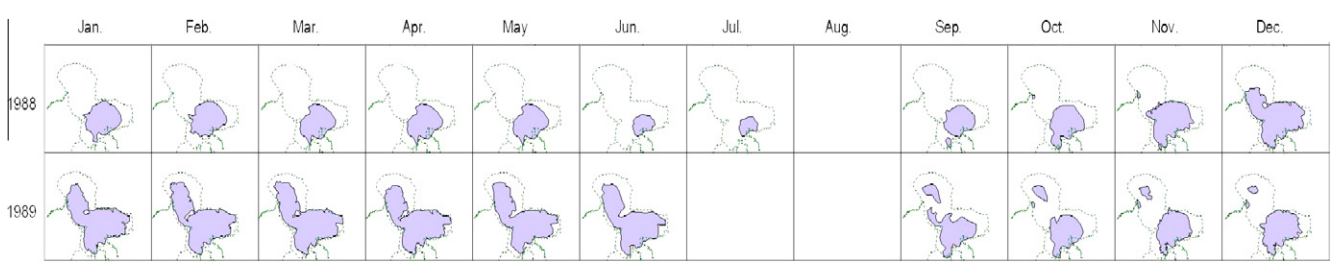

Fig. 7. Example of monthly fluctuations of the total inundated area of Lake Chad (open water and water under aquatic vegetation) as derived from Meteosat Tmax data.

The entire lake, only exceeded 15,000 km^2 (60% of the lake 25,000 km^2 total area as defined by the 284 m amsl shoreline) for only 12% of the study period. That is, from 1986 to 2001, the entire lake remained below 60% of its maximum 284 m amsl shoreline extent for ~88% of the time.

Since the low inundation extents in 1990, increasing inter-annual trends are apparent for both the northern pool and the entire lake. The linear trends for the 1990–2000 period indicate increasing rates of ~750 km^2/yr for the entire lake, and ~500 km^2/yr for the northern pool. Most (68%) of the annual expansion in the extent of the entire Lake Chad (7800 km^2) observed during this 90–00 decade has contributed to the northern pool (5300 km^2).

Fig. 9 synthesizes the space and time patterns of the inundation of Lake Chad in a map showing the percentage of time each Meteosat Tmax pixel is inundated during the study period (1987–2001). A large permanent water body is clearly apparent in front of the Chari delta (highlighted in red) and known to be mostly composed of a vast open water area surrounded by a ring of dense aquatic vegetation. The occurrence of water in the northern pool is frequent in a large channel linking the southern to the northern pool across the Great Barrier. It leads to a low lying area in the southwest part of the northern pool where water becomes gradually more scarce to totally absent in a ~5–10 km buffer zone around the eastern 284 m amsl shoreline and in a ~20–40 km buffer zone around the north and west 284 m amsl shoreline (Fig. 9). A small area in front of the delta of the Komadugu Yobe (north of Malam Fatori, Fig. 1) is frequently inundated.

The inundation area of Lake Chad is known to be strongly linked to the flow of the Chari River and the rainfall over its basin (Olivry et al., 1996). Annual inflows from the Chari River below ~16 km^3/yr fail to generate any inundation in the northern pool (observed in 86/87; 87/88; 89/90; 90/91 and 93/94); the seasonal increase in

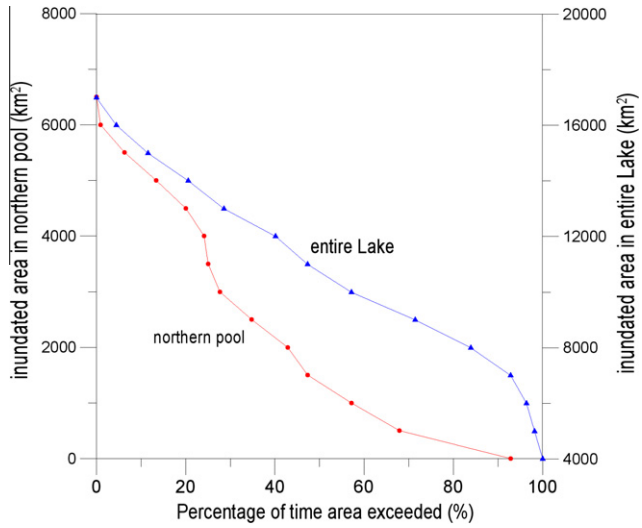

Fig. 8. Inundated area duration curves for the northern pool (red) and the entire Lake Chad (blue).

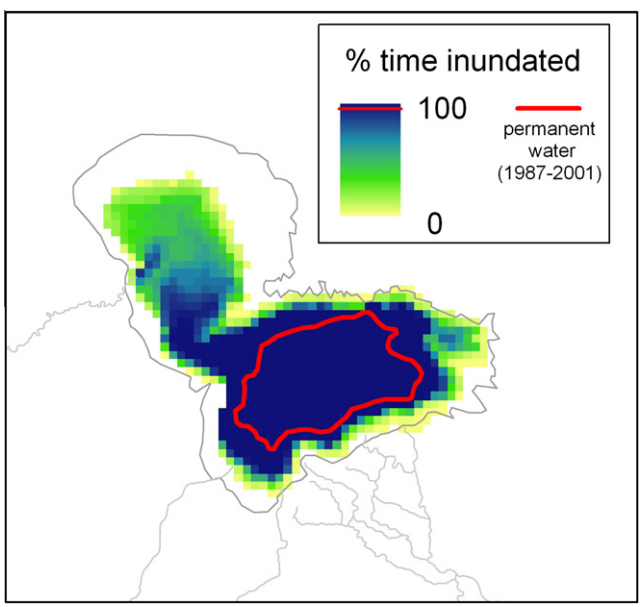

Fig. 9. Map showing the monthly inundation patterns (October–June) of Lake Chad during the drought period (1987–2001) including open water and water under aquatic vegetation in terms of percentage of time inundated.

the entire lake extent for these hydrological years is constrained to the south pool (Fig. 6). The 97/98 Chari inflow which had a monthly peak flow of only 1400 m^3/s but a total annual flow of 18 km^3 generated a ~1600 km^2 increase in the lake area; while in comparison the 89/90 and 93/94 inflows which had a higher monthly peak flow of 1590 m^3/s and 1460 m^3/s respectively but a smaller total annual flow of only 16 and 15 km^3 did not generate any significant increase in the lake area.

Tmax observations confirm that the extent of the inundated area in the 'Small' Lake Chad is strongly dependant on the flow of the Chari (Lemoalle, 1991). It is possible to build a simple model (R^2 = 0.78; Root Mean Square Error (RMSE) = 947 km^2) of the annual maximum inundated areas in the northern pool for the 'Small' Lake Chad (*Smax*) using the following simple linear regression:

$$Smax(i) = \mathrm{Max}(0; 8.97^{*}Q(i-1) - 3098) \quad (1)$$

With i, the year; $Q(i)$ the average discharge ($m^3\,s^{-1}$) of the Chari from March of the year i to February of the year $i+1$; $Smax(i)$ the maximum inundated area (km^2) in the northern pool between January and May of the year i.

This relationship can be further improved by taking into account the maximum inundated area of the previous year ($Smax(i-1)$) using the following multiple linear regression:

$$Smax(i) = \mathrm{Max}(0; 8.814^{*}Q(i-1) + 0.3297^{*}Smax(i-1) - 3914) \quad (2)$$

Fig. 10 shows the good agreement between the predicted maximum inundation area of a 'Small' Lake Chad using Eq. (2) and the observed inundation area using Meteosat Tmax (R^2 = 0.86; RMSE = 756 km^2) (see Fig. 11).

Based on rainfall data published by Olivry et al. (1996) and analysed using Thiessen's method, the mean annual rainfall over the entire lake for the period 1971–1989 is of 223 mm (197 mm in the northern basin). As a first approximation, the annual input of direct rainfall on the lake can be estimated for each year of the period 1986–2000 by multiplying the area of the lake derived from Meteosat Tmax data by the mean annual rainfall for the 1971–1989 period. The results obtained show that, depending on the year of the period 1986–2000, rainfall represents between 6% and 14% of the total annual inflow to the lake.

5. Discussion

5.1. Advantages, limitations and future developments of Meteosat Tmax

To supplement some of the limitations of the radar, microwave and optical data, the thermal methodology described here can be used operationally for the monitoring of the total inundated area of large water bodies covered by aquatic vegetation, and is relevant for semi-arid and arid regions. The thermal data from the geosynchronous satellites (e.g. Meteosat, GOES, GMS) are particularly pertinent in that respect in tropical regions thanks to their long archive and high temporal resolution, allowing to bypass clouds. A main limitation of the Meteosat Tmax method remains its inability to differentiate surface water from wet soils which restricts its applicability to low rainfall conditions. Nonetheless, a number of large lakes and wetlands in arid and semi-arid regions have their maximum inundation outside of the local wet season; and the Tmax method described here could yield useful hydrological information (e.g., Yaeres, Niger and Okavango inland deltas, and Sudd in Africa, and Lake Eyre Basin in Australia).

Continued research efforts are required to further improve our capacity to map and monitor surface water from space. The Meteosat series of satellites which operated since 1977 has been replaced by a new generation called Meteosat Second Generation (MSG) in 2002. MSG has the potential to provide continuity and improvement to the Meteosat-derived Tmax data presented in this paper. Specifically, MSG offers greater temporal (imaging repeat cycle of 15 min instead of 30 min for Meteosat) resolution and spatial (Thermal data at 2.5 km at nadir instead of 5 km for Meteosat) (Schmetz et al., 2002). Amongst the most promising potential developments in remote sensing of surface water is the future Surface Water and Ocean Topography (SWOT) mission. It is currently planned to be launched between 2013 and 2016 and will provide significant improvements in our abilities to map inundated areas from space (http://www.swot.jpl.nasa.gov/mission/). Using wide-swath altimetry technology, SWOT will provide temporal and spatial variations in water volumes stored in rivers, lakes, and wetlands at unprecedented resolution (Biancamaria et al., 2010). SWOT will generate a global 3D mapping of all terrestrial water bodies

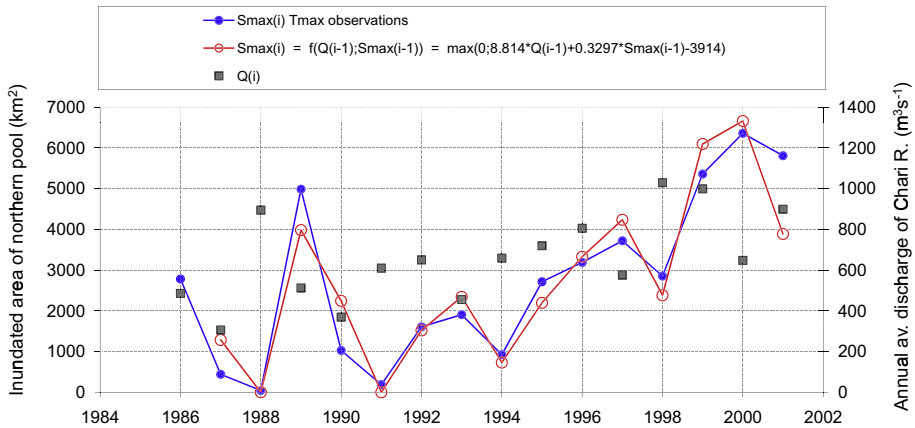

Fig. 10. Observation and modelling of the annual maximum inundated area in the northern pool. The blue line represents the observed annual maximum inundation from Meteosat Tmax data. The red line is the maximum inundation area modelled from a multiple linear regression using the annual maximum inundated extent from the previous year and the discharge of the Chari-Baguirmi from March to February.

Fig. 11. Photos (a and b) showing the return of Lake Chad in the northern pool as observed in June 1999 – vicinity of Doro fishing village 14°04′48″N–13°29′46″E (Leblanc, 2002). (c) An example of the dense aquatic vegetation characteristic of the 'Small' Lake Chad (photo by Lemoalle and Quensière, 1989).

whose surface area exceeds 250 m^2 and rivers whose width exceeds 100 m (Biancamaria et al., 2010). The principal instrument of SWOT will be a Ka-band Radar Interferometer (KaRIN), which will provide heights and co-registered all weather imagery of water over 2 swaths, each 60 km wide, with an expected precision of 1 cm/km for water slopes, and absolute height level precision of 10 cm/km^2.

5.2. Global hydrological change

Drying of lakes and wetlands has now become a global environmental change phenomenon with many striking examples from around the world (Arnel, 2002; UNEP, 2005). In particular closed lakes, that are extensively distributed in arid and semi-arid regions of the world, are highly sensitive to human impacts and climate variability and change (Nihoul et al., 2004). Once the fourth largest inland water body, since 1960 the Aral Sea level has fallen 23 m, its area has shrunk 74%, its volume has decreased 90%, and its salinity has risen from 10 to more than 100 g/l (Micklin, 2007). The level of the Dead Sea has dropped by about 17 m between 1977 and 2000 (Gertman and Hecht, 2002; Gravrieli and Oren, 2004). In Australia, the severe ongoing drought (Leblanc et al., 2009) exacerbated by water diversions has caused water levels at Lake Corangamite, Australia's largest permanent saline lake, to drop by ~2.3 m and the inundated area to shrink by 50% since 1993; in the same area a simultaneous drying is also observed in many smaller closed lakes (Tweed et al., 2009). The main causes for the drying of these closed lakes are climate variability and change (small closed lakes of the Corangamite region), water diversions/consumption (Aral Sea; Dead Sea) or both (Lake Corangamite).

For Lake Chad, reasons for the decline from an 'Average' to a 'Small' lake must be re-examined. Coe and Foley (2001) have reported that about half of the decrease in area from an 'Average' to a 'Small' Lake Chad since the 1970s can be attributed to a significant expansion of irrigation in the Chari and Komadugu Yobe catchment (four fold irrigation increase from 2.5 km^3/yr in 1953–1979 to 11.2 km^3/yr in 1983–1994 with 90% of the irrigation occurring in the Chari catchment). These irrigation estimates originate from a personal communication of which the details remain unpublished. Lemoalle (2004) reviewed irrigation infrastructure capacity and found that current irrigation in the Chari system only amounts to 0.2 km^3/yr; and therefore cannot be a dominant cause of the decrease from an 'Average' to a 'Small' Lake Chad. Irrigation from the lake water itself also seems insignificant relative to the overall lake's water budget. The largest irrigation scheme (polders) in the eastern archipelago of Lake Chad only uses 75 Mm^3/yr (0.075 km^3/yr) (Lemoalle, 2004). Instead the decrease in Lake Chad area can be predominantly attributed to climate variability and a decline in rainfall in the Chari basin (Lemoalle, 2004). Except during 'Large' Lake Chad periods, Lake Chad is a closed lake. River and rainfall inputs are offset by evaporation losses, some seepage and by volume and area fluctuations according both to seasonal and annual variations of the inflows. As these inflows are closely associated with rainfall over the basin, the lake is highly sensitive to climate variability. In natural conditions, it is a good indicator of climate fluctuations. Rainfall follows the displacement of the Inter Tropical Convergence Zone (ITCZ) which moves northward over the active part of the basin from February to August. Since the beginning of the 1970s, the ITCZ has not been moving as far north as before resulting in lower rainfall over the basin (L'Hôte et al., 2002). A long period of above average rainfall from 1950 to 1967 over the West African Sahel was followed by below average rainy seasons after 1970, with the succession of harmful droughts that peaked in 1972, 1973 and 1984 (L'Hôte et al., 2002). Over the Chari-Logone Basin, annual rainfall has decreased by about

150 mm from 1095 mm/yr (1950–1967) to 932 mm/yr (1972–2006) with a north–south displacement of the rainfall gradient of 150 km (data from Climate Research Unit available at www.cru.uea.ac.uk). A ±10% change in rainfall over the basin results in a ±30% change in the annual discharge of the Chari River, the main tributary to the lake, which has also been lowered as follows: period 1960–1969: 41.0 km^3; 1970–1979: 31.1 km^3; 1980–1989: 16.6 km^3; 1990–1999: 20.4 km^3; 2000–2006: 22.0 km^3 (Fig. 2a). This rainfall deficit is the main reason for the decrease in Lake Chad level and the shrinking of surface area from an 'Average' to a 'Small' lake (Lemoalle, 2004; Bader et al., 2011).

5.3. The 'Small' Lake Chad replaced in the context of past fluctuations

The contemporary 'Small' Lake Chad described in this paper must be replaced in the broader context of past fluctuations. Over the past centuries and millennia, Lake Chad has repeatedly dried and filled under the control of natural climate variability. Olivry et al. (1996) compiled observed and simulated information on the fluctuations of Lake Chad since 1860s. A 'Large' Lake Chad, with a water level around 284 m amsl has been described by some European explorers who visited the lake in the 19th century: Denham in 1822, Barth and Overweg in 1851, Rohlfs in 1866, Nachtigal in 1870 and Monteil in 1892. But they also reported from local tradition that a 'Small' Lake Chad probably occurred at some time between 1822 and 1850 (Carmouze et al., 1983). Later military missions indicated that the lake had receded between 1904 and 1915 into a 'Small' Lake Chad with large areas of previously open waters emerging as marshes or dry land (Destenave, 1903a,b; Freydenberg, 1907; Tilho, 1910; Garde, 1911). From 1920 to 1940 an 'Average' Lake Chad occurred with water level ~281.5 m. Water level dropped rapidly from 1940 to 1942 and stayed relatively low (min ~280 m) until 1946. The return of a wetter period is then observed and the lake level rose from 1946 to 1965 to reach a maximum of ~283.3 m in January 1963, 1964 and 1965. Since 1965, the protracted ongoing drought induced a sharp decline in water level which led to the partition of the northern and southern pools in 1973 and the 'Small' Lake Chad still observed today (Olivry et al., 1996). Maley (1981, 2004) reconstructed the fluctuations of Lake Chad during the past millennium using palynology, ^{14}C datings and events from the regional history. A total of six 'Small' Lake Chad episodes may have occurred in the last millennium. In addition to the three episodes presented above (mid 19th century, beginning of 20th century and present) another three 'Small' Lake Chad have probably occurred in the middle of the 15th, 16th and 18th centuries; with a possible complete drying of Lake Chad in the 15th century (Maley, 1981, 2004). Periods of 'Large' Lake Chad are recorded between these dry phases. The existence of a much larger lake during the late Quaternary, referred to as Megalake Chad, is now well established (Tilho, 1925; Pias and Guichard, 1957; Schneider, 1967; Maley, 1981; Servant and Servant, 1983; Schuster et al., 2003, 2005; Leblanc et al., 2006a,b). The last Holocene Megalake Chad filling occurred between ca. 8500 and 6300 calendar years before present (cal. yr BP), with three major sub-phases: 8500–8200, 7900–7400 and 6900–6300 cal. yr BP (Maley, 2004; Leblanc et al., 2006b). At its maximum extent, Megalake Chad had an area of about 340,000 km^2 (only 8% less than the present-day world's largest lake, the Caspian Sea), a volume of 13,500 km^3 and a maximum depth of about 160 m (Leblanc et al., 2006a,b). In comparison, at a water level of 282 m, the 'Average' Lake Chad has a mean depth of 3.5 m, a total area of 20 000 km^2 and a volume of ~70 km^3 (Bader et al., 2011).

5.4. Socio-economical value of a 'Small' Lake Chad

Wetlands provide globally significant social, economic, and environmental benefits (Costanza et al., 1997). The drought in the African Sahel, which started with the low rainfall of 1972 and 1973, has pushed populations toward those areas where water allowed for cattle breeding, cultivation and fishing. As Lake Chad retreated, local population adapted to the new environment; and the 'Small' Lake Chad still provides immense socio-economical services to the region. Recession cultivation has largely benefitted from the increased seasonal variations in water level and area in the 'Small' Lake Chad compared with the 'Average' Lake Chad. This is especially true in the southern basin where the inundation/recession cycle has been fairly reproducible from year to year. In the northern basin, while the natural resources for cultivation, herding and fishing are abundant in wet years, when part of the basin is inundated, the populations have to migrate during dry years when no inundation occurs (Africa and Lemoalle, 1996; Sarch, 1996, 2001; Sarch and Birkett, 2000).

6. Conclusion

Aquatic vegetation now covers most of the 'Small' Lake Chad and must be accounted for in estimates of the total inundated area. For example in June 1999, the water covered by aquatic vegetation represented ~75% of the total inundated area. We propose a new technique that takes advantage of the high temporal resolution and thermal sensor from geosynchronous satellite to enable the mapping of both open water and water covered by aquatic vegetation in large water bodies. The Meteosat Tmax technique was applied successfully to Lake Chad but should also work for the inundation monitoring of large lakes, wetlands and floodplains in other semi-arid regions using Meteosat and MSG data (e.g. Yaeres, Niger inland delta, Okavango inland delta, Sudd) and other weather satellites such as GOES and MTSAT (e.g., floodplains of the Lake Eyre Basin in Australia and Andean Altiplano Lakes in South America).

The annual flow of the Chari River, which is influenced by rainfall patterns in the wet tropics and equatorial southern regions of the basin, controls the extent of both northern and southern pools of Lake Chad. The increasing trend observed from 1990 to 2001 is only a partial recovery from the peak of the drought – the entire study period (1986–2001) remains that of a 'Small' Lake Chad with a maximum inundated area only briefly reaching ~16,000 km^2 in 2000. However, the very strong response of the lake to more humid conditions as observed in this study in 1988/1989 shows that the lake could recover very quickly from the drought in the event of a return to higher rainfall in the Chari basin.

Our satellite observations provide a comprehensive and consistent time series of Lake Chad total area, which have been used to develop a hydrological model that includes the functioning of the 'Small' Lake Chad (Bader et al., 2011). Such a model would be valuable as a predictive tool in water management, for instance to forecast the annual inundation of Lake Chad or to simulate the impact of the proposed interbasin transfer from the Ubangi river (Congo basin).

Acknowledgments

Discharge data on the Chari River at N'Djamena are from the Direction des Ressources en Eau et de la Météorologie (DREM, Chad). Meteosat thermal composite data (Tmax) are from the "Veille Climatique Satellitaire" programme (1985–2001). The authors particularly wish to extend their thanks Dr. Dominique Dagorne from the IRD for providing the Meteosat Tmax data and for his invaluable insight on their hydrological application.

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